THE ALGEBRA OF GLOBAL FUNCTIONS OF A PROPER SCHEME OVER A FIELD AND ITS CONNECTED COMPONENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The last two parts are obtained by the author's explicit 'of course (5) implies (6)' and 'of course (7) implies (8)' steps; both sentences are reproduced as written.

Editorial note. The counted spans carry 7 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-proper-geometrically-reduced-global-sections is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Cohomology of a proper scheme over an affine base is finite. Cohomology of Schemes, Lemma coherent-lemma-proper-over-affine-cohomology-finite, invoked at proof lines 1954-1956. Supplies that A is a finite dimensional k-algebra.
- A finite dimensional algebra over a field is a product of local Artinian algebras. Algebra, Lemma algebra-lemma-finite-dimensional-algebra and Proposition algebra-proposition-dimension-zero-ring, invoked at proof lines 1960-1962. Supplies the product decomposition in part (2).
- Flat base change for the cohomology of the structure sheaf. Cohomology of Schemes, Lemma coherent-lemma-flat-base-change-cohomology, invoked at proof lines 1982-1983 and 1992-1993. Supplies A tensor k-bar = H^0 of the base changed scheme, used in parts (4), (5) and (7).
- Characterization of separable field extensions and of geometrically reduced purely inseparable extensions. Algebra, Lemmas algebra-lemma-characterize-separable-field-extensions and algebra-lemma-geometrically-reduced-finite-purely-inseparable-extension, invoked at proof lines 1986-1988. Supplies the separability conclusion of part (4).

Source reference. Original lemma 030W: sources/stacks-chow/tags.txt, source lines 3963–3963. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 00J6: sources/stacks-chow/tags.txt, source lines 704–704. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 030V: sources/stacks-chow/tags.txt, source lines 3962–3962. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00EU: sources/stacks-chow/tags.txt, source lines 548–548. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original proposition 00KJ: sources/stacks-chow/tags.txt, source lines 753–753. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02O6: sources/stacks-chow/tags.txt, source lines 3505–3505. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k.*

1. *$A = H^0(X, \mathcal{O}_X)$ is a finite dimensional k-algebra,*
2. *$A = \prod_{i=1,\ldots,n} A_i$ is a product of Artinian local k-algebras, one factor for each connected component of X,*
3. *if X is reduced, then $A = \prod_{i=1,\ldots,n} k_i$ is a product of fields, each a finite extension of k,*
4. *if X is geometrically reduced, then k_i is finite separable over k,*
5. *if X is geometrically connected, then A is geometrically irreducible over k,*
6. *if X is geometrically irreducible, then A is geometrically irreducible over k,*
7. *if X is geometrically reduced and geometrically connected, then $A = k$, and*
8. *if X is geometrically integral, then $A = k$.*

Proof. By Cohomology of Schemes, Lemma 02O6 we see that $A = H^0(X, \mathcal{O}_X)$ is a finite dimensional k-algebra. This proves (1).

Then A is a product of local Artinian k-algebras by Algebra, Lemma 00J6 and Proposition 00KJ. If $X = Y \amalg Z$ with Y and Z open in X, then we obtain an idempotent $e \in A$ by taking the section of $\mathcal{O}_X$ which is 1 on Y and 0 on Z. Conversely, if $e \in A$ is an idempotent, then we get a corresponding decomposition of X. Finally, as X has a Noetherian underlying topological space its connected components are open. Hence the connected components of X correspond 1-to-1 with primitive idempotents of A. This proves (2).

If X is reduced, then A is reduced. Hence the local rings $A_i = k_i$ are reduced and therefore fields (for example by Algebra, Lemma 00EU). This proves (3).

If X is geometrically reduced, then $A \otimes_k \overline{k} = H^0(X_{\overline{k}}, \mathcal{O}_{X_{\overline{k}}})$ (equality by Cohomology of Schemes, Lemma 02KH) is reduced. This implies that $k_i \otimes_k \overline{k}$ is a product of

fields and hence k_i/k is separable for example by Algebra, Lemmas 030W and 030V. This proves (4).

If X is geometrically connected, then $A \otimes_k \overline{k} = H^0(X_{\overline{k}}, \mathcal{O}_{X_{\overline{k}}})$ is a zero dimensional local ring by part (2) and hence its spectrum has one point, in particular it is irreducible. Thus A is geometrically irreducible. This proves (5). Of course (5) implies (6).

If X is geometrically reduced and geometrically connected, then $A = k_1$ is a field and the extension k_1/k is finite separable and geometrically irreducible. However, then $k_1 \otimes_k \overline{k}$ is a product of $[k_1 : k]$ copies of $\overline{k}$ and we conclude that $k_1 = k$. This proves (7). Of course (7) implies (8). □